

Optimal Planning of Distribution Systems and Charging Stations Considering PV-Grid-EV Transactions

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Abstract—Uncertainties associated with large-scale deployment of electric vehicles (EVs) and photovoltaic (PV) pose challenges to distribution network expansion planning (DNEP). This paper proposes an optimal planning method for EV charging stations (EVCS) and distribution systems to accommodate the ever-increasing uncertainties. It is achieved by entailing PV-grid-EV transactions, which enables EVCS and PV prosumers to trade energy to make profits while complying with grid securities. It offers a cost-effective operational alternative to long-term planning, which would otherwise result in PV curtailment or unnecessary DNEP. The transactive market operates on a peer-to-peer (P2P) basis and is cleared via a decentralized algorithm to protect privacy and enable autonomous decision-making. EVCS is incentivized by a designed network charge, quantifying its impact on adhering to security constraints from both long-term and short-term perspectives. Considering EV users' charging decisions, we derive optimal EV charging prices to regulate EV charging flow. We employ multiple linearization techniques to ensure the convergence of the non-convex model. Results demonstrate that the proposed method enables the distribution networks to accommodate the large-scale integration of EV and PV more effectively.

Index Terms—Distribution network expansion planning, EV charging station, network charges, PV-grid-EV transactions, PV prosumer.

NOMENCLATURE

Indices and Sets

a, Ω_{PV}	Index and set of alternatives for PV.
b, Ω_{SS}	Index and set of alternatives for substations.
c, Ω_{EV}	Index and set of EV chargers.
e, Ω_{ES}	Index and set of alternatives for energy storage (ES).
g, G	Index and set of roads in the transportation networks.

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ij, Ω_I	Index and set of nodes in the distribution networks.
$ij/l, L$	Index and set of lines in the distribution networks.
$n/m, \Omega_P$	Index and set of all prosumers and EVCS (peer).
$h/q, \Omega_R$	Index and set of nodes in the transportation networks.
s, S	Index and set of scenarios.
t, T	Index and set of operation/ transactive interval.
u, Ω_{CS}	Index and set of substations.
v, V	Index and set of EV.
w, Ω_{AL}	Index and set of alternatives for upgrading lines.
y	Index of planning interval, year.
Ξ, Θ	& Set of prosumers and EVCS participated in PV-grid-EV transactive market.

Parameters

C	Investment costs of lines, PV, EVCS, EV chargers, ES, and substation.
$E_{s,n,0}^{FL/ES}$	Initial energy of flexible load and ES.
$\bar{E}_n^{ES}$	Maximum energy of ES.
$\bar{I}_l$	Maximum line current.
L_l	Length of line l .
N^I, N^S, N^P	Total number of existing nodes, substations, and peers.
$\bar{P}_{n,t}^{DG}$	Maximum unit output power of PV.
$\bar{P}_{n,t}^{FL}, \underline{P}_{n,t}^{FL}$	Minimum/ maximum range of flexible load.
$\bar{P}_{n,t}^{ES, ch/dis}$	Maximum charge/ discharge power of ES.
P_c^{EV}	Rated charging power of EV charger c .
$SOC_{n,v}^{EV,d}$	Pre-determined desired SoC of EV v in EVCS n .
$V_{min}^{ES, ch/dis}, V_{max}^{ES, ch/dis}$	Minimum/ maximum nodal voltage.
η_n	ES charging/discharging efficiency.
ζ	EV consumption ratio.

Variables

$g_{l,w}, b_{l,w}$	Conductance, susceptance of line l .
$I_{s,ij,t}$	Current magnitude squared of line ij .
$N_{n,c,y}^{EV}$	Number of EV charger c constructed in EVCS n .
$P_{s,n,t}^{DSO}$	Guaranteed trading power with main grid.

$P_{s,n,t}^{\text{ES,ch/dis}}$	Charging/discharging power of ES in EVCS n .
$P_{s,n,c,t}^{\text{EV,ch/dis}}$	Charging/discharging power of EV charger c .
$P_{s,n,t}^{\text{FL/I}}$	FL and response power in prosumer n .
$P_{s,u,t}^{\text{M}}$	Main-grid exchange power at substation u .
$P_{s,n,t}^{\text{T}}$	Transactive power of peer n .
$P_{s,ij,t}, Q_{s,ij,t}$	Active/reactive power of line ij .
$r_{ij,w}/x_{ij,w}$	Resistance/reactance of line ij .
$S_{u/l,y}^{\text{P}}$	Expanded capacity of line l and substation u .
$S_{n,y}^{\text{DG/ES}}$	Expanded capacity of PV and ES.
$S_{l,t}^{\text{P,sqr}}, S_{u,t}^{\text{P,sqr}}$	Apparent power squared of line l and substation u .
$S_{n,y}^{\text{EVCS}}$	Expanded capacity of EVCS.
$\text{SOC}_{s,n,v}^{\text{EV}}$	SoC level of EV v in EVCS n .
$T_{n,v}^{\text{CS}}$	Charging time of EV v in EVCS n .
$V_{s,i,t}/V_{s,j,t}$	Voltage magnitude squared at node i/j .
$\alpha_{l,w,y}^{\text{P}}$	Updated status of line l with line alternative w .
$\beta_{u,b,y}^{\text{P}}$	Updated status of substation u with alternative b .
$\sigma_{n,a,y}^{\text{DG}}$	Updated status of PV with alternative a .
$\gamma_{n,e,y}^{\text{ES}}$	Updated status of ES with alternative e .
$u_{l,y}^{\text{P}}$	Binary variable associated with a line investment (1 if line is constructed; 0 otherwise).
$\lambda_{s,n,t}^{\text{I/T}}$	Energy interactive and transactive price of peer n .
$\varphi_{n,y}^{\text{EVCS/EV}}$	Updated status of EVCS, EV chargers of EVCS n .
$\phi_{s,n,t}^{\text{I/O}}$	Investment/ operation network charges of peer n .
$\pi_{s,n,t}^{\text{I/O}}$	EV charging price of EVCS n .
$\theta_{l,t}^{\text{P}}$	Voltage angle difference across line l .

I. INTRODUCTION

THE INTEGRATION of electric vehicles (EVs) and photovoltaic (PV) holds great potential for reducing carbon emissions in distribution and transportation networks. EVs on low-carbon energy supplied by PV are a promising alternative to traditional fossil-fueled vehicles. However, the proliferation of EVs supported by PV poses various challenges to the operation and planning of distribution systems.

The proliferation of EVs relies on EV charging stations (EVCS). The lack of EVCS will result in dissatisfaction among EV users. The planning of EVCSs is widely studied in the existing literature [3], [4], [5], [6], [7]. The primary tenant of EVCS planning is to determine the optimal location and capacity of EVCS to benefit the economy of EVCS and the convenience of EV users [3], [4], [5]. Concerning the negative impact of uncertain EV charging load on the distribution networks, the power loss reduction, voltage support and reliability improvement of the distribution networks are considered the objectives of the EVCS planning problems [6], [7]. However, the above literature addressed the EVCS planning problem, assuming that the distribution networks have sufficient capacity to accommodate the futuristic growth of EV

charging load and neglected the need for distribution network expansion planning (DNEP).

The EVCS planning should complement DNEP to cope with the ever-increasing EV charging load. Moreover, the uncertain charging load may seriously jeopardize network operation and advance the need for DNEP [8]. On the other hand, it is pointed out in [9] that the planning scheme of EVCS and the charging scheduling of EVs can influence the spatial and temporal distribution of EV charging load and help alleviate potential network security issues caused by PV. Thus, the co-optimization of EVCS planning and DNEP has become an essential topic and has been investigated widely [10], [11], [12], [13], [14]. For instance, the optimal EVCS siting and sizing decisions are co-optimized with the DNEP in [10], aiming to minimize the total investment and operation costs. The planning elements to be decided in the DNEP problem generally include the substations, lines, PV generators, and capacity banks [11].

Although the studies above explored the joint planning of EVCSs and distribution systems, the following challenges persist. Firstly, the entities involved in the joint planning should include EVCSs, PV prosumers, EV users, and the independent distribution system operator (IDSO) [14]. Each entity has different objectives, e.g., network secure operation for IDSO, self-profit for EVCSs and PV prosumers, and charging expenditure for EV users, which lead to conflicts [15]. The traditional centralized optimization approach can not accurately characterize the competitive relationships. Secondly, while EVs supported by PV can reduce carbon emissions, its uncertain output requires additional DNEP, which would otherwise result in energy imbalance, over-voltage, and frequency fluctuation [16]. In the PV-EV integrated networks, placing exclusive reliance on long-term planning leads to suboptimal utilities regarding uncertainties originating from PV generation and EV charging load.

These challenges can be addressed by entailing short-term energy transactions between PV and EV (hereinafter referred to as PV-grid-EV transactions) with long-term planning. Existing studies have validated the potential of PV-grid-EV transactions in reducing PV curtailment [17], increasing EVCS's profits [18], and reducing EV charging expenditure [19]. PV-grid-EV transactions can enable the optimal allocation of EV charging flexibility and PV generation in centralized [20] and decentralized market mechanisms [21]. It is essential to find the equilibrium of profit-driven EVCS decisions rather than directly maximizing the overall profit to establish a transparent, competitive market and boost the motivation of EVCSs for providing EV flexibility to the networks. Incorporating PV-grid-EV transactions in the distribution networks provides alternatives to mitigate excessive dispatch on EVs. It enables the co-optimization of the power flow and EV charging flow coupling by the EVCSs [22]. Our previous work demonstrated that energy transactions among prosumers could reduce reliance on upstream generation, which would otherwise result in PV curtailment or unnecessary DNEP while disregarding the charging behaviors of EVs [23]. However, all the existing literature on PV-grid-EV transactions is operation-oriented, and their application in the

joint planning of EVCSs and distribution systems has not been investigated.

One key challenge in implementing PV-grid-EV transactions to facilitate the planning of EVCSs and DNEP is to coordinate EV charging behaviors of profit-driven EVCS and enhance the operational flexibility of the networks. In the competitive market, EVCS decisions are driven by profit considerations and may not prioritize network security [24]. In this regard, EV charging flexibility and PV-grid-EV transactions should be guided by appropriate monetary incentives to adhere to network security and defer DNEP. Existing EV monetary incentives are directly determined by the IDSO [25], [26] or the EVCS [27] for maximizing their payoffs. In that case, the optimal EV charging prices are either spatial-temporal fixed or decided by profit-driven IDSO and EVCS [28]. That existing EV monetary incentive design does not rigorously quantify the planning and operation costs of EVCSs and distribution networks incurred by EV charging behaviors, creating wrong guidance for EV charging flows. Only through the accurate quantification and allocation of the incurred costs can the EV charging flexibilities be fully used fairly and transparently.

To fill the research gap, we propose a joint planning model of distribution systems and EVCSs complemented by PV-grid-EV transactions to accommodate the integration of EVs and RES. The solution provides the optimal planning decisions for substations, distribution lines, PV generation, ES, and EV chargers. Unlike existing research, we coordinate the planning elements in the PV-grid-EV transactions market to enhance the operational flexibility of the networks and defer DNEP while improving EVCS's profits and lowering EV users' expenditures. The primary contribution of this paper is as follows:

- 1) The PV-grid-EV transactions are integrated into the joint planning problem of distribution systems and EVCSs. It provides EVCSs, PV prosumers, and the distribution networks with short-term operational flexibility rather than relying exclusively on long-term planning, thus making planning more flexible and efficient.

- 2) A PV-grid-EV transactions algorithm is proposed, enabling the co-optimization of power flow and EV charging flow. It replaces the extra dispatch on EV charging flow with energy transactions through the incentive of EV charging price, which reduces EV user's charging distance, increases EVCS's profit, and enables the seamless integration of PV.

- 3) An EV charging price is proposed to incentivize EV charging behaviors and enable a secure and economic network. It is cost-reflective as it quantifies the incurred planning and operation costs related to EVCSs and networks. It leads to an optimal planning scheme regarding EV satisfaction level and EVCS profits while adhering to network security.

The remainder of the paper is structured as follows. Section II gives the overall design of the joint planning model. Section III describes the model formulation for EVCS, PV prosumers and the distribution networks. The monetary incentives for EVCSs and EV users are designed in Section IV, along with the EV traffic assignment model. Section V develops the solution methodology for the joint planning problem in the transactive market. Simulation results are

analyzed in Section VI. Finally, conclusions are summarized in Section VII.

II. OVERALL FRAMEWORK

In this section, we first describe the overall framework of the proposed model and then discuss its coordination with PV-grid-EV transactions toward secure economic networks.

A. Stakeholder Characterization

The stakeholders involved are classified as IDSO, EVCSs, EV users, and PV prosumers. EVs are accessible to the distribution networks through EVCSs with EV charging prices. Responding to the spatial-temporary charging prices designed by EVCSs, EV users actively adjust their charging routes and plans to minimize their total charging expenditures. EVCSs are profit-driven entities that decide the planning of EV chargers to support EV charging load. EVCSs leverage EV charging flexibility to participate in PV-grid-EV transactions to maximize their payoff while satisfying EV users' charging plans. PV prosumers, equipped with PVs, energy storages (ESs) and flexible loads, can decide the PV-grid-EV transactions and investment of PVs and ESs.

On the other hand, IDSO owns the network assets and is accountable for facilitating the integration of PV and EV. IDSO also carries out the expansion planning and operation of the networks. With PV-grid-EV transactions involving EVCSs and PV prosumers, IDSO should establish suitable network charges that incentivize PV prosumers, EVCSs, and EV users to make decisions that align with the grid-friendliness objectives.

B. Coordinated Relationship Between Joint Planning and PV-Grid-EV Transactions

The PV-grid-EV transactive market operates on a peer-to-peer (P2P) basis, enabling EVCSs and PV prosumers to trade energy directly through multi-bilateral negotiations. PV-grid-EV transactions offer a cost-effective alternative to meet the EV charging load instead of relying solely on the grid. When considering the impact of traffic flow on EV users' satisfaction, PV-grid-EV transactions replace the need for additional grid dispatch for EV charging with energy transactions, resulting in reduced charging costs for EV users. This, in turn, influences the optimal placement and sizing of EVCS and PV prosumers.

From IDSO's perspective, PV-grid-EV transactions are valuable for leveraging EV charging and PV prosumer generation flexibility. This approach helps address energy imbalances and alleviate network congestion, providing operational flexibilities as a short-term complement to long-term expansion planning. It enables the seamless integration of intermittent EV charging loads and PV generation while deferring unnecessary DNEP.

C. Framework of Joint Planning With PV-Grid-EV Transactions

We model the optimal planning problem as a bi-layer decision-making process. In the upper-layer, IDSO undertakes the expansion planning and operation of the network

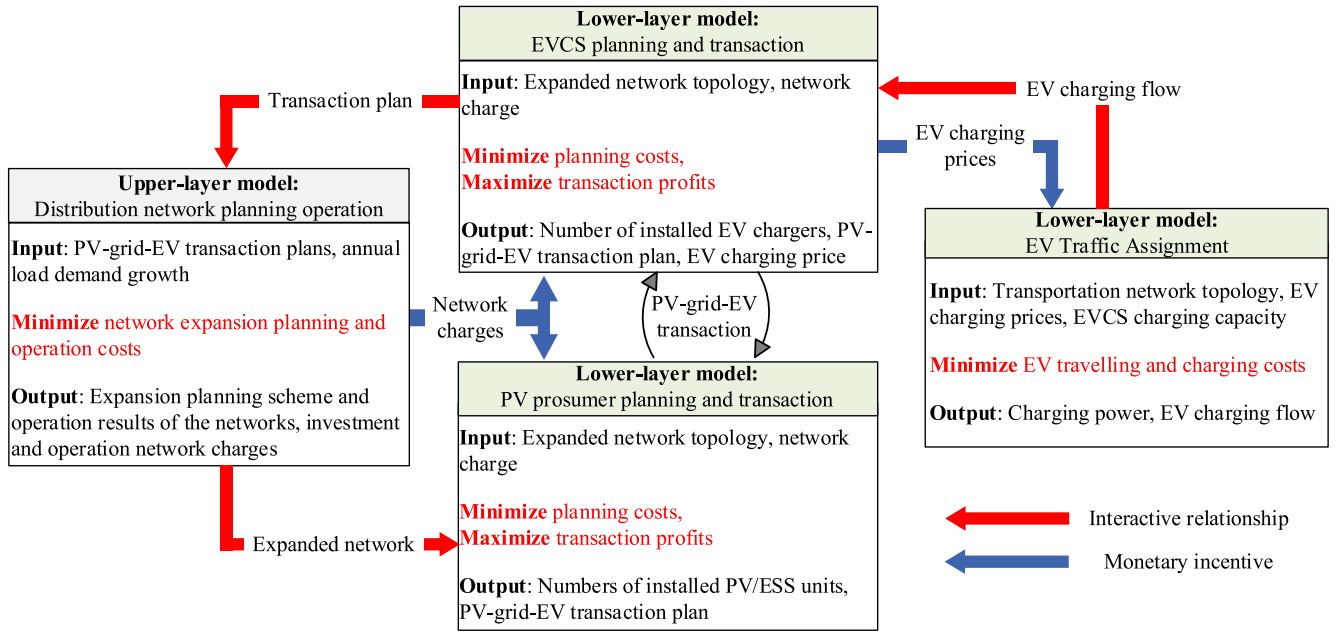


Fig. 1. Framework for the distribution network expansion planning model in the PV-grid-EV transactive market.

to achieve cost-effectiveness. It's noted that since PV-grid-EV transactions rely on the physical grids owned by the IDSO, EVCS decisions impose additional requirements on the DNEP. The planning and operation costs associated with the distribution networks due to PV-grid-EV transactions should be allocated based on EVCS's responsibilities. With the presence of EVCSs, the IDSO formulates the optimal network charges to incentivize the planning, scheduling, and transaction patterns of EVCSs on a grid-wide scale. The framework for the distribution network expansion planning model in the PV-grid-EV transactive market is shown in Fig. 1.

The lower-layer determines the optimal planning scheme for EVCSs and PV prosumers in the transactive market. The lower-layer model encompasses autonomous decision-making processes by EVCSs and PV prosumers. In this process, EVCSs make decisions regarding their planning and participation in PV-grid-EV transactions with PV prosumers to cover network charges and provide feedback to the IDSO. From the perspective of EVCS, the network charges incurred result from EV charging demands and should be distributed to EV users through EV charging prices.

III. JOINT PLANNING MODEL FORMULATION

This section describes the main features of the joint planning model of distribution systems and EVCSs, formulated as a bi-layer decision-making process. In the upper-layer model, IDSO decides the expansion planning scheme and the operation strategy of the distribution networks, respectively. The third-layer model involves an autonomous decision-making process by EVCS and PV prosumers in the transactive market.

A. Upper-Layer IDSO Expansion Planning Model

In the upper-layer model, IDSO is responsible for identifying the optimal location and capacity for lines and substations

and finding the optimal operation strategy concerning PV-grid-EV energy transaction plans [15]. The objective of upper-layer is to accommodate the PV-grid-EV transactions demand with the lowest possible investment and operation costs. Accordingly, the upper-layer IDSO model is expressed as (1).

$$\min C_y^P = \left(\underbrace{\sum_{w \in \Omega_{AL}} \sum_{l \in L} \alpha_{l,w,y}^P C_w^{CL} L_l^P S_{l,y}^L}_{\text{Cost of investing lines}} + \underbrace{\sum_{s \in S} \sum_{t \in T} \sum_{l \in L} C_{l,t}^O \cdot P_{s,l,t} + \sum_{u \in \Omega_{CS}} \sum_{b \in \Omega_{SS}} \sigma_{u,b,y}^P C_b^{SS} S_{u,y}^S}_{\text{Cost of maintenance lines}} \right) / (1 + \tau)^{t-1} \quad (1)$$

$$\text{s.t. } \sum_{w \in \Omega_{AL}} \alpha_{l,w,y}^P = 1, \sum_{c \in \Omega_{SS}} \sigma_{u,c,y}^P = 1 \quad u \in \Omega_{CS}, l \in L \quad (2)$$

$$(P_{s,l,t})^2 + (Q_{s,l,t})^2 = S_{s,l,t}^{P,sqr} \leq \sum_{w \in \Omega_{AL}} \alpha_{l,w,y}^P \cdot (S_{l,y}^P)^2 \quad (3)$$

$$P_{s,l,t} = \sum_{w \in \Omega_{AL}} \alpha_{l,w,y}^P \cdot \left(\frac{(V_{s,i,t})^2 g_{l,w} - V_{s,i,t} V_{s,j,t} (b_{l,w} \sin \theta_{s,l,t} + g_{l,w} \cos \theta_{s,l,t})}{V_{s,i,t} V_{s,j,t} (b_{l,w} \cos \theta_{s,l,t} - g_{l,w} \sin \theta_{s,l,t})} \right) \quad (4)$$

$$Q_{l,t}^P = \sum_{w \in \Omega_{AL}} \alpha_{l,w,y}^P \cdot \left(\frac{-(V_{s,i,t})^2 b_{l,w} + V_{s,i,t} V_{s,j,t} (b_{l,w} \cos \theta_{s,l,t} - g_{l,w} \sin \theta_{s,l,t})}{V_{s,i,t} V_{s,j,t} (b_{l,w} \sin \theta_{s,l,t} + g_{l,w} \cos \theta_{s,l,t})} \right) \quad (5)$$

$$(P_{s,u,t})^2 + (Q_{s,u,t})^2 = S_{s,u,t}^{P,sqr} \leq \sum_{b \in \Omega_{SS}} \sigma_{u,b,y}^P \cdot (S_{u,y}^P)^2 \quad u \in \Omega_{CS} \quad (6)$$

$$V_{\min} \leq V_{s,i,t} \leq V_{\max}, \quad 0 \leq I_{s,ij,t} \leq I_{ij,\max} \quad (7)$$

$$\sum_{l \in L} v_{l,y}^P = N^I - N^S \quad (8)$$

$$\sum_{j \in L} k_{ji} - \sum_{ij \in L} k_{ij} = K^i \forall i \in \Omega_I \quad (9)$$

$$K^i = 1 \forall i \in \Omega_P, K^i = 0 \forall i \notin \Omega_P \cup \Omega_{CS} \quad (10)$$

$$|k_{ij}| \leq N^P v_{l,y}^P \forall ij \in L \quad (11)$$

Eq. (2) is the investment decision constraint, ensuring that for each expanded line and substation, only one alternative can be executed each year. The binary variable of the selected

alternative changes from 0 to 1, while the binary variable of other alternatives remains 0. Eq. (3) ensures that the power flow in each line does not exceed the upgrading line capacity. The power flow model is constrained in (4)–(5). Eq. (6) indicates that the power purchased from the main grid should not exceed the substation capacity. Eq. (7) ensures that operation variables are within the security constraints. Eqs. (8)–(11) ensures the radiality of the expanded distribution networks, preventing any loop in the expanded networks, based on the definition of the graph tree [29]. The fictitious load of PV prosumers is denoted by K^i , which should be powered by the substation, and k_{ij} represents the fictitious flow associated with line ij .

B. Lower-Layer EVCS Planning Model

The stakeholders involved in the lower-layer model include the EVCSs and PV prosumers. EVCSs autonomously decide on the planning of EV chargers and engage in PV-grid-EV transactions in the competitive market. The decision-making problem for EVCS is formulated in (12)–(18).

$$\begin{aligned} \max R_{n,y}^{\text{EVCS}} = & \sum_{c \in \Omega_{\text{EV}}} \sum_{s \in S} \sum_{t \in T_{n,y}^{\text{CS}}} \left(\eta_n^{\text{EV, ch}} P_{s,n,c,t}^{\text{EV, ch}} - P_{s,n,c,t}^{\text{EV, dis}} / \eta_n^{\text{EV, dis}} \right) \pi_{s,n,t}^{\text{EV}} - \\ & \underbrace{\sum_{s \in S} \sum_{t \in T} P_{s,n,t}^{\text{T}} \lambda_{s,n,t}^{\text{T}}}_{\text{Revenue of P2P transaction}} + \underbrace{\sum_{m \in \Xi^{-n}} \sum_{s \in S} \sum_{t \in T} (\phi_{s,n,t}^{\text{O}} + \phi_{s,n,t}^{\text{I}})}_{\text{Network charges}} - \underbrace{\sum_{s \in S} \sum_{t \in T} P_{s,n,t}^{\text{M}} \lambda_{s,n,t}^{\text{M}}}_{\text{Cost of guaranteed trading}} \\ & - \underbrace{\left(\varphi_{n,y}^{\text{EVCS}} C_{n,y}^{\text{EVCS}} + \sum_{c \in \Omega_{\text{EV}}} \varphi_{n,y}^{\text{EV}} C_c^{\text{EV}} N_{n,c,y}^{\text{EV}} P_c^{\text{EV}} \right)}_{\text{Cost of constructing EV chargers}} \end{aligned} \quad (12)$$

$$\text{s.t. } \varphi_{n,y}^{\text{EVCS}} \geq \varphi_{n,y-1}^{\text{EVCS}}, \varphi_{n,y}^{\text{EV}} \geq \varphi_{n,y-1}^{\text{EV}} \quad (13)$$

$$\varphi_{n,y}^{\text{EV}} \leq \varphi_{n,y}^{\text{EVCS}} \quad (14)$$

$$0 < P_{s,n,v,t}^{\text{EV, ch}} < \bar{P}_{n,v}^{\text{EV, ch}}, 0 < P_{s,n,v,t}^{\text{EV, dis}} < \bar{P}_{n,v}^{\text{EV, dis}} \quad (15)$$

$$\text{SOC}_{n,v}^{\text{EV, d}} \leq \text{SOC}_{n,v}^{\text{EV}} + \sum_{t \in T_{n,y}^{\text{CS}}} \left(\eta_n^{\text{EV, ch}} P_{s,n,c,t}^{\text{EV, ch}} - \frac{P_{s,n,c,t}^{\text{EV, dis}}}{\eta_n^{\text{EV, dis}}} \right) < \text{SOC}_{n,v}^{\text{EV, max}} \quad (16)$$

$$\sum_{c \in \Omega_{\text{EV}}} P_c^{\text{EV}} N_{n,c,y}^{\text{EV}} \Delta t \geq \sum_{t \in T_{n,y}^{\text{CS}}} \sum_v (\text{SOC}_{n,v}^{\text{EV}} - \text{SOC}_{n,v,0}^{\text{EV}}) \quad (17)$$

$$\forall \text{EV arrived during time interval } t \quad (18)$$

$$T_{n,v}^{\text{CS}} = \frac{(\text{SOC}_{n,v}^{\text{EV}} - \text{SOC}_{n,v,0}^{\text{EV}})}{P_c^{\text{EV}}} \quad (18)$$

Eq. (13) is the investment decision constraint, indicating that EVCS can only be constructed once at each candidate node over the planning horizon, and the planning of EV chargers is irreversible. Eq. (14) ensures that EV chargers are not permitted to be constructed until the EVCS has been placed at the node. Eq. (15) ensures that EVs are with secure operation constraints. Eq. (16) is the EV charging/discharging time-coupling constraint, ensuring that the SOC of EV v at departure time must be greater than or equal to the predetermined desired SOC, which implies that the total delivered energy during the whole period must satisfy the requirement of EV users. Eq. (17) establishes that the energy that can be supplied by the constructed EV chargers in EVCS n should satisfy the overall charging demand required by EVs arrived

at each time interval. Eq. (18) calculates the charging time of EV v at EVCS n .

C. Lower-Layer PV Prosumer Planning Model

PV prosumers coordinate RES planning, demand response, and bilateral energy transactions with EVCSs. The planning considerations for PV prosumers involve PV units and ES units. Besides the conventional load, flexible load within the PV prosumer enables the active demand response. Serving as additional generation sources in the distribution networks, PV prosumers leverage their resources in both the long term and the short term to maximize their interests. The decision-making problem for PV prosumers is formulated in (19).

$$\begin{aligned} \max C_{n,y}^{\text{PR}} = & \underbrace{\sum_{s \in S} \sum_{t \in T} (P_{s,n,t}^{\text{I}} \lambda_{s,n,t}^{\text{I}})}_{\text{Revenue of demand response}} + \underbrace{\sum_{s \in S} \sum_{t \in T} (P_{s,n,t}^{\text{T}} \lambda_{s,n,t}^{\text{T}})}_{\text{Revenue of energy trading}} - \\ & \underbrace{\sum_{m \in \Xi^{-n}} \sum_{s \in S} \sum_{t \in T} (\phi_{s,n,t}^{\text{O}} + \phi_{s,n,t}^{\text{I}})}_{\text{Network charges}} - \underbrace{\sum_{a \in \Omega_{\text{DG}}} \sigma_{n,a,y}^{\text{DG}} C_{n,a}^{\text{DG}} S_{n,y}^{\text{DG}}}_{\text{Cost of constructing PV generator}} \\ & - \underbrace{\sum_{s \in S} \sum_{t \in T} (P_{s,n,t}^{\text{M}} \lambda_{s,n,t}^{\text{M}})}_{\text{Cost of guaranteed trading}} - \underbrace{\sum_{e \in \Omega_{\text{ES}}} \gamma_{n,e,y}^{\text{ES}} C_{n,e}^{\text{ES}} S_{n,y}^{\text{ES}}}_{\text{Cost of constructing ES}} \end{aligned} \quad (19)$$

$$\text{s.t. } \sum_{a \in \Omega_{\text{DG}}} \sigma_{n,a,y}^{\text{DG}} = 1, \sum_{e \in \Omega_{\text{ES}}} \gamma_{n,e,y}^{\text{ES}} = 1 \quad (20)$$

$$0 < P_{s,n,t}^{\text{DG}} < \bar{P}_{s,n,t}^{\text{DG}}, \underline{P}_{s,n,t}^{\text{FL}} < P_{s,n,t}^{\text{FL}} < \bar{P}_{s,n,t}^{\text{FL}} \quad (21)$$

$$0 < P_{s,n,t}^{\text{ES, ch}} < \bar{P}_{s,n,t}^{\text{ES, ch}}, 0 < P_{s,n,t}^{\text{ES, dis}} < \bar{P}_{s,n,t}^{\text{ES, dis}} \quad (22)$$

$$0 < E_{s,n,0}^{\text{ES}} + \sum_{t=1}^T \left(\eta_n^{\text{ES, ch}} P_{s,n,t}^{\text{ES, ch}} - P_{s,n,t}^{\text{ES, dis}} / \eta_n^{\text{ES, dis}} \right) < \bar{E}_n^{\text{ES}} \quad (23)$$

Eq. (20) is the investment decision constraint, which indicates that only one alternative for DG and ES can be constructed for each prosumer. Eqs. (21)–(22) ensure that DG, ES, and flexible load have secure operation constraints. Eq. (23) is the ES time-coupling constraint.

IV. INCENTIVE MECHANISMS FOR PROFIT-DRIVEN EVCS AND EV

EVCSs and EV users do not inherently prioritize grid-friendliness over their profitability, which could potentially compromise network security. In a competitive market, profit-driven EVCSs and EV users can only effectively coordinate through economic incentives. Taking into account the interactive relationships between the distribution networks, EVCSs, and EV users, the monetary incentive should be established in two steps: 1) The network charges designed by the IDSO for EVCSs, which quantify the grid-related planning and operation costs associated with the decisions made by EVCSs; 2) The charging prices designed by EVCSs for EV users, which quantify the planning and operation costs incurred by EV users' decisions related to the EVCSs.

It is noted that the design of monetary incentive should consider the benefits of three entities: 1) the IDSO, which aims at adhering to network security constraints at the lowest possible costs; 2) the EVCSs, which aims at maximizing its profits; and 3) EV users who aim at minimizing its

travel expenses (EV charging costs, equivalent time cost). Inappropriate monetary incentives could lead to incorrect guidance for EVCSs, potentially compromising network operation, or they may not be sufficient to recover the costs incurred by EVCS decisions. From the perspective of both EVCSs and EV users, inappropriate incentives could favor network operations at the expense of EVCS profits or EV user satisfaction. This conflict among the three entities necessitates assigning corresponding responsibilities through monetary incentives. Finally, we propose a traffic assignment model to capture EV users' response behaviors to the designed monetary incentives.

A. Network Charges for EVCSs

The network charge for EVCSs comprises two components considering the grid-related costs incurred in different stages: 1) investment network charge for recovering long-term investment costs; 2) operation network charge for recovering short-term operation costs.

During the expansion planning, IDSO decides on network reinforcements to ensure the secure operation of EVCSs. As PV-grid-EV transactions depend on lines and guaranteed transactions with IDSO depend on substations, the investment network charge allocated to each EVCS should cover investment costs for both lines and substations. To define the portion of the transactions from the investment costs incurred in each line, we adopt the AC Power Transfer Distribution factor (AC-PTDF) to track the corresponding power flowing through line l due to transactions nm . The AC-PTDF is calculated based on the AC OPF model of the distribution networks. Investment costs for substations should be assigned to those EVCSs that engage in guaranteed trading with IDSO. In addition, the operation network charge is influenced by P2P transactions and the operational state of the distribution networks. In this regard, we utilize DLMPs to encourage PV-grid-EV transactions that support network operations and discourage unfavorable transactions from the perspective of IDSO [30].

In this regard, given the AC-PTDF and DLMP, the optimal network charge designed for EVCS n is expressed as:

$$\phi_{s,n,t}^I = \frac{1}{2} \sum_{s \in S} \sum_{l \in L_L} C_w^{\text{CL}} L_l S_{l,y}^{\text{L}} \frac{P_{s,nm,l,t}^{\text{T}}}{P_{s,l,t}} + \sum_{u \in \Omega_{\text{CS}}} C_b^{\text{CS}} S_{u,y}^{\text{S}} \frac{P_{s,n,u,t}}{P_{s,u,t}} \quad (24)$$

$$\phi_{s,n,t}^O = \frac{1}{2} (\lambda_{s,n,t}^{\text{DLMP}} - \lambda_{s,m,t}^{\text{DLMP}}) P_{s,nm,t}^{\text{T}} \quad (25)$$

where $P_{s,nm,l,t}^{\text{T}}$ is the active power of line l due to transactions nm . Notably, the incurred investment costs for lines and operation costs are equally shared between the seller and buyer.

Due to substantial economies of scale in investments, the main purpose of peers paying the investment network charge is to reduce the incurred congestion costs, which are considered part of the operation network charge. Financial transmission rights (FTR) are risk-hedging instruments designed to mitigate congestion cost risks in the contractual market. FTRs grant holders the right to earn a profit based on differences in DLMPs resulting from network congestion. Those who pay an

investment network charge for grid development are entitled to a corresponding FTR. Real-time transactions within this range are exempt from congestion charges. However, in some adverse scenarios where FTR is insufficient, peers have no choice but to either reduce excessive trading power or pay a higher operation network charge, which is modified as follows.

$$\phi_{s,n,t}^O = \frac{1}{2} (\lambda_{s,n,t} - \lambda_{s,m,t}) \cdot \max[(P_{s,nm,t}^{\text{T}} - P_{nm,t}^{\text{T}*}), 0] \quad (26)$$

where $P_{nm,t}^{\text{T}*}$ is the pre-purchased transactive energy capacity.

B. Optimal Charging Prices for EV

EV charging prices are monetary tariffs formulated by EVCSs to recover the planning and operation costs of EVCSs and the incurred network charges. The optimal EV charging prices internalize the planning and operation status of the distribution networks and regulate EV charging flow to facilitate the operation flexibilities of the networks and defer long-term DNEP. The EV charging prices are expressed as:

$$\pi_{s,n}^I = \varphi_{n,y}^{\text{EVCS}} C^{\text{EVCS}} S_n^{\text{EVCS}} + \varphi_{n,y}^{\text{EV}} C^{\text{EV}} S_{n,y}^{\text{EV}} + \sum_{s \in S} \sum_{t \in T} \phi_{s,n,t}^I / S_{n,y}^{\text{EV}} T \quad (27)$$

$$\pi_{s,n,t}^O = \sum_{s \in S} \sum_{t \in T} \phi_{s,n,t}^O / \sum_{c \in \Omega_{\text{EV}}} (\eta_n^{\text{EV, ch}} P_{s,n,c,t}^{\text{EV, ch}} - P_{s,n,c,t}^{\text{EV, dis}} / \eta_n^{\text{EV, dis}}) \quad (28)$$

$$\pi_{s,n,t}^{\text{EV}} = \pi_{s,n}^I + \pi_{s,n,t}^O \quad (29)$$

The EV charging prices should recover both investment and operation costs. The investment-oriented charging prices $\pi_{s,n}^I$ aim to recover the investment costs of distribution systems and EVCSs. The operation-oriented charging prices $\pi_{s,n,t}^O$ allocate operation network charges to EV users. Notably, $\pi_{s,n,t}^O$ is spatial-temporal differential, which can regulate EV charging flow to align with the constraints of distribution and transportation networks.

C. Traffic Assignment Model for EV

This section introduces the user equilibrium model [25] to capture EV users' responses to the charging prices designed in Section IV-A.

$$\min \sum_{f_{p,r}} \sum_{g \in G} \omega \int_0^{x_g} t_g(h) dh + \sum_{n \in N} \omega \int_0^{x_n} t_n(h) dh + \sum_{r \in R} \sum_{p \in \tilde{p}^r} \omega t_{p,r}^{\text{ch}} f_{p,r} + \sum_{n \in N} \pi_{s,n,t}^{\text{EV}} F_n^{p,r} \quad (30)$$

$$\text{s.t. } x_g = \sum_{r \in R} \sum_{p \in \tilde{p}^w} f_{p,r} \delta_{p,r,g}, \quad \forall g \in G, x_n = \sum_{r \in R} \sum_{p \in \tilde{p}^w} f_{p,r} v_{p,r,g} \quad (31)$$

$$t_g(x_g) = t_g^0 (1 + 0.15(x_g/c_g)^4), t_n(x_n) = t_n^0 \left[1 + J \frac{x_n}{c_n - x_n} \right] \quad (32)$$

$$\sum_{p \in \tilde{p}^r} f_{p,r} = d_r, \quad \forall r \in R \quad (33)$$

$$f_{p,r} \geq 0, \quad \forall p, r \quad (34)$$

$$p_n^{\text{dc}} = \sum_{r \in R} \sum_{p \in \tilde{p}^r} f_{p,r} F_n^{p,r} \quad (35)$$

$$SOC_q^r - SOC_h^r + d_g \zeta - F_h^{p,r} = \rho_g^r, \quad \forall (q, h) = g \in G \quad (36)$$

$$-M(1 - z_g^r) \leq \rho_g^r \leq M(1 - z_g^r), \quad \forall g \in G \quad (37)$$

where ω is the value of the unit time of the EV user, x indicates the traffic flow, and is the charging energy of path p between O-D pair r at EVCS n . z_g^r indicates whether the path traverses link g . t_g and t_n indicate the travel time on a link g and EVCS n with a traffic flow x_g and x_n . t_g^0 and t_n^0 are the free traveling time and charging time. c_g and c_n are the travel capacity of link g and the maximum allowable charging flow of EVCS n . The relationship between the path flow and the flow on link g and that which culminates at EVCS n are expressed as (31). The traffic congestion level on link g and at EVCS n are indicated in (32). Eq. (33) ensures that the sum of path flows should equal the corresponding OD demand variable. The nonnegative path flows are ensured by (34). The total charging load at EVCS n is calculated in (35). The relationship between the SoC level and the charging choice of EVs in the transportation networks is expressed as (36)–(37).

V. SOLUTION METHODOLOGY

To leverage the advantages of the P2P transactions for protecting privacy and enabling autonomous decision-making, we use the Alternating Direction Method of Multipliers (ADMM) algorithm for solving the joint planning problem within the PV-grid-EV transactive market. The bi-layer model formulates a mixed-integer nonlinear programming (MINLP) problem containing integer planning variables. The convergence of a decentralized market clearing algorithm is not guaranteed due to the nonconvex MINLP problem. Thus, we adopt an alternating optimization procedure (AOP) and several linearization techniques, such as big-M method and piecewise linearization, to ensure convergence of the proposed model.

A. Decentralized Market Clearing Algorithm by ADMM

The concept of the ADMM algorithm is compatible with the characteristics of P2P energy transactive market for enabling autonomous decision-making and protecting privacy. Based on the EVCS problem (12), the decentralized market-clearing problem for PV-grid-EV transactions is expressed as (38).

$$\begin{aligned} \min & C_n^{\text{Inv}} + C_n^{\text{Ope}} + C_n^{\text{EV}} - C_n^{\text{Gt}} + \sum_{m \in \Xi^{-n}} \sum_{s \in S} \sum_{t \in T} (\phi_{s,n,t}^0 + \phi_{s,n,t}^1) \\ & + \underbrace{\sum_{s \in S} \sum_{t \in T} \sum_{m \in \Xi_n} \left(\lambda_{s,nm,t}^{T(k)} \left(\frac{P_{s,nm,t}^{T(k)}}{2} - P_{s,nm,t}^{T(k)} \right) + \frac{\rho}{2} \left(\frac{P_{s,nm,t}^{T(k)} - P_{s,nm,t}^{T(k)}}{2} - P_{s,nm,t}^{T(k)} \right)^2 \right)}_{\text{P2P Negotiation Mechanism of PV-grid-EV Transaction}} \quad (38) \end{aligned}$$

where k is the iteration number, ρ is the penalty factor, $\rho > 0$. $\lambda_{s,nm,t}^{T(k)}$ is the dual variable of consensus constraint, also defined as the P2P energy transactive price. In each iteration, the dual variables are updated via (39).

$$\lambda_{s,nm,t}^{T(k+1)} = \lambda_{s,nm,t}^{T(k)} - \rho \left(P_{s,nm,t}^{T(k+1)} - P_{s,nm,t}^{T(k+1)} \right) / 2 \quad (39)$$

The procedure of the ADMM-based decentralized market clearing algorithm is as follows: 1) EVCS and prosumers

optimize (12) and (19) in parallel, coordinating its decisions related to planning and PV-grid-EV transactions; 2) EVCS and prosumers share trading-related variables. $P_{s,nm,t}^T$ with its trading peers. 3) Upon receipt of the updated transaction information, EVCS and prosumers update $\lambda_{s,nm,t}^T$ and calculate the local residuals; 4) IDSO summarizes local residuals broadcasted by EVCS and prosumers to test if the global residual meets the stopping criteria, expressed as (40). The above iteration is repeated until convergence.

$$\sum_{n \in N} \sum_{m \in \Xi_n} \left(\left(P_{s,nm,t}^{T(k+1)} - P_{s,nm,t}^{T(k)} \right)^2 + \left(P_{s,nm,t}^{T(k+1)} + P_{s,nm,t}^{T(k)} \right)^2 \right) \leq \varepsilon \quad (40)$$

B. Convergence Analysis of ADMM Algorithm for Non-Convex Problem (38)

Despite ADMM being proven efficient and accurate in solving convex problems, its convergence cannot be guaranteed for non-convex problems [31]. As defined in (12)–(18), decision variables related to EVCS planning are discrete. Such discrete variables result in a nonconvex and discontinuous market clearing model (38), for which the standard ADMM does not guarantee convergence. To ensure convergence, we embed the AOP within ADMM. The main principle of AOP is to convert the problem, which is solved by ADMM, into continuous and convex. Thus, we reformulate (35) in a compact form.

$$\min_{\chi, \eta} \left\{ C^{\text{EVCS}}(\chi, \eta) : \text{s.t. } \chi \in \Omega^\chi, \eta \in \Omega^\eta \right\} \quad (41)$$

where $C^{\text{EVCS}}(\cdot)$ denotes the objective function (38). χ and Ω^χ are the set and feasible regions of continuous decision variables where $\chi = [P_{s,n,c,t}^{\text{EV, ch}}, P_{s,n,c,t}^{\text{EV, dis}}, P_{s,n,t}^{\text{M}}, P_{s,nm,t}^{\text{T}}, P_{s,n,t}^{\text{PV}}]$. η and Ω^η are the set and feasible regions of discontinuous decision variables where $\eta = [\varphi_{n,y}^{\text{EVCS}}, \varphi_{n,y}^{\text{EV}}, g_n^{\text{EVCS}}, g_n^{\text{EV}}]$. It is noted that χ is the set of variables related to operation while η is the set of variables related to the planning. The detailed steps of AOP embedded with ADMM are as follows:

1) *Solve the Operation-Related Decision-Making Problem:* Prosumers determine their optimal transactive plan with planning-related decision variables remaining fixed. Minimize $C^{\text{EVCS}}(\chi, \eta)$ over $\chi \in \Omega^\chi$ with fixed $\eta = \eta^k$ and obtain χ^k .

$$\chi^k = \arg \min_{\chi \in \Omega^\chi} C^{\text{EVCS}}(\chi, \eta^k) \quad (42)$$

2) *Solve the Planning-Related Decision-Making Problem:* EVCS determines their optimal planning scheme with operation-related decision variables remaining fixed. Minimize $C^{\text{EVCS}}(\chi, \eta)$ over $\eta \in \Omega^\eta$ with fixed $\chi = \chi^k$ and obtain η^{k+1} .

$$\eta^{k+1} = \arg \min_{\eta \in \Omega^\eta} C^{\text{EVCS}}(\chi^k, \eta) \quad (43)$$

Then, χ^k and η^{k+1} have the following properties:

$$\begin{cases} C^{\text{EVCS}}(\chi^k, \eta^k) \leq C^{\text{EVCS}}(\chi^k, \eta^k), & \forall \chi \in \Omega^\chi \\ C^{\text{EVCS}}(\chi^k, \eta^{k+1}) \leq C^{\text{EVCS}}(\chi^k, \eta), & \forall \eta \in \Omega^\eta \end{cases} \quad (44)$$

By (44), we have $C^{\text{EVCS}}(\chi^{k+1}, \eta^{k+1}) \leq C^{\text{EVCS}}(\chi^k, \eta^k)$, thus $\{C^{\text{EVCS}}(\chi^k, \eta^k)\}$ is a nonincreasing sequence. Since (χ^k, η^k) are feasible solutions of (41), the

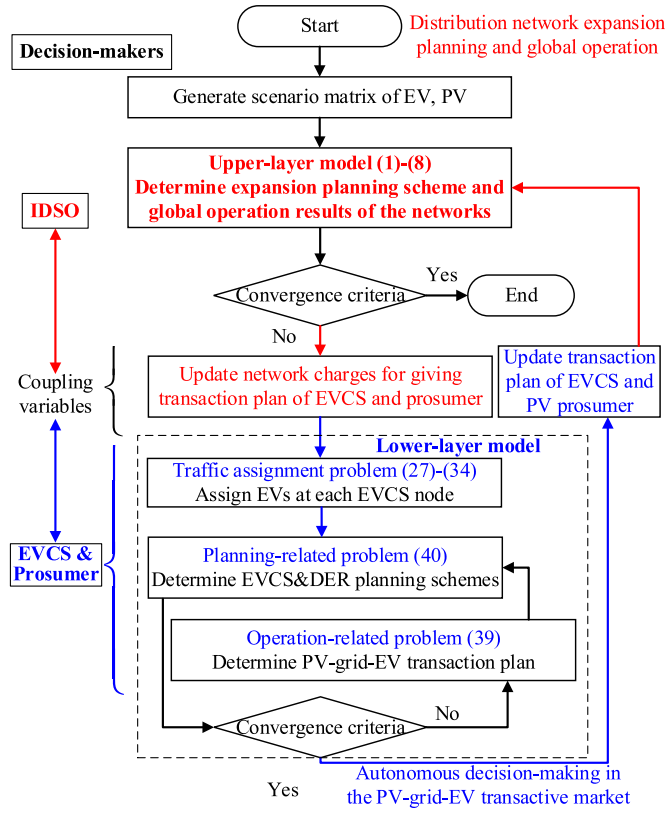


Fig. 2. Schematic flowchart of the proposed joint planning method.

values of $C^{EVCS}(\chi^k, \eta^k)$ are bounded below by the optimal value of (41). Below bounded nonincreasing sequence $\{C^{EVCS}(\chi^k, \eta^k)\}$ must converge.

C. Linearization Process of the DNEP Model

The upper-layer problem is nonconvex because the product of two decision variables in (3)–(6), including line capacity constraints (3), power flow constraints (4)–(5), and substation capacity constraints (6), is nonlinear. We utilize several linearization techniques, such as big-M method and piecewise linearization, to linearize these nonlinear constraints. The full approximation procedures can be found in the Appendix [33].

The schematic flowchart of the proposed joint planning model with PV-grid-EV transactions is depicted in Fig. 2.

VI. CASE STUDIES

A. Case Description

The efficiency of the proposed model is validated on the coupled IEEE 33-node distribution system [30] and 10-node transportation system [16]. Furthermore, its scalability is assessed in an integrated 40-node transportation system and 72-node power distribution system derived from data from a real-world city in central China. The proposed problem has been simulated for 12 typical daily scenarios yearly during the five-year planning horizon. The time granularity of planning is one year. Technical specifications of EVs are taken from [32]. The rated charging power for each EV charger is 44kW, and the charging efficiency is 92%. Data from the planning alternative and uncertainty model are presented in

TABLE I
CASE DESCRIPTIONS

Case	Expansion planning	PV-grid-EV transactions	Monetary incentives
1	✓	×	×
2	✓	✓	×
3	✓	✓	✓

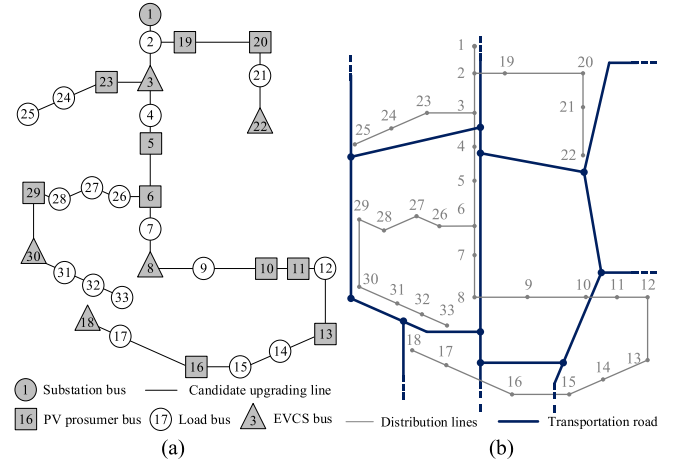


Fig. 3. Topology of the coupled (a) IEEE 33-node distribution system and (b) 10-node transportation system.

the Appendix [33]. The investment cost for each EVCS is assumed to be \$163,000, while the investment cost for each EV charger is \$23,500 [32].

The outcome of this work lies in the cost-efficient joint planning of EVCS and distribution systems complemented by virtual trading. In this section, we evaluate the impact of the proposed method on distribution systems planning by designing three cases, as shown in Table I.

B. Expansion Planning Results in the 33-Node Distribution System and 10-Node Transportation System

This section discusses the numerical results for each test case using the coupled IEEE 33-node distribution system and 10-node transportation system, as shown in Fig. 3. Among the 33 nodes in the distribution system, 10 are PV prosumers equipped with PV and ES, and 5 are EVCSs. In comparison, the other 17 nodes only have conventional loads.

Table II compares the optimal expansion planning schemes of the distribution networks and costs of IDSO in three cases. Tables III and IV summarize the optimal planning schemes and operation results of EVCSs and PV prosumers. Notably, the results of the PV prosumers in Case 1 are not included in Table IV, as this case does not account for the planning and operation of PV and ES. Additionally, Fig. 4 provides worst-case line loading in the distribution networks in Case 3. Table II and Fig. 4 show that the loading percentages in lines 1-2, 2-3, 3-4, 6-7, and 6-26 are the highest in Case 3, indicating that these lines suffer from severe network congestion. This further necessitates the expansion planning for these lines.

1) *The Impact of PV-grid-EV Transactions on the Planning of EVCS and Distribution System by Comparing Cases 1*

TABLE II
OPTIMAL PLANNING SCHEMES AND COSTS OF IDSO IN THREE CASES

Case	Planning scheme		Network charge (10 ⁴ \$)	Overall cost (10 ⁴ \$)
	Line	Substation		
1	1-2(VII), 2-3(VI), 3-4(V), 4-5(V), 5-6(V), 6-7(IV), 7-8(II), 6-26(II)	+4MW	/	679.42 (0%)
	1-2(VII), 2-3(VI), 3-4(V), 6-7(IV), 7-8(II)	+2 MW	/	580.91 (-14.5%)
	1-2(VI), 2-3(V), 3-4(IV), 6-7(II), 6-26(II)	+1 MW	46.37	532.67 (-21.6%)

TABLE III
OPTIMAL PLANNING SCHEMES AND COSTS OF EVCSs IN THREE CASES

EVCS (node)	Case	Planning scheme (in chargers)	Average waiting time (min)	Total revenue (10 ⁴ \$)
1(3)	1	9	14.79	53.04 (0%)
	2	14	13.65	60.84 (+14.7%)
	3	12	11.28	66.14 (+24.7%)
2(8)	1	11	16.73	64.73 (0%)
	2	15	14.78	72.76 (+12.4%)
	3	13	12.83	80.02 (+23.6%)
3(18)	1	7	11.26	42.54 (0%)
	2	10	10.58	46.62 (+9.6%)
	3	10	10.17	51.69 (+21.5%)
4(22)	1	7	10.15	49.93 (0%)
	2	9	8.09	54.27 (+8.7%)
	3	11	8.26	62.96 (+26.1%)
5(30)	1	6	9.73	39.43 (0%)
	2	7	7.79	43.49 (+10.3%)
	3	10	8.12	49.96 (+26.8%)

and 3: The results in Table II indicate that, compared to Case 1, the expansion of lines 4–5, 5–6, and 7–8 are deferred in Case 3. Additionally, the capacity expansions of other lines and substations have also decreased. Consequently, this leads to a decrease in the total costs of the IDSO by 21.6%, which can be attributed to the energy transactions among EVCSs and PV prosumers. In Case 1, where PV-grid-EV transactions are excluded, the IDSO meets the EV charging demand from the substation and several main distribution lines. This energy supply pattern can lead to network congestion in main lines during peak hours, necessitating unnecessary additional capacity for substations and expansion planning of lines. In Case 3, which incorporates PV-grid-EV transactions, EVCSs can procure energy from PV prosumers instead of relying solely on grid power. This approach alleviates the burden on substations and main lines, leading to a more evenly balanced power flow distribution throughout the networks.

Additionally, as shown in Table III, compared with Case 1, in which PV-grid-EV transactions are disregarded, the total revenue of EVCS in Case 3 has been improved to varying degrees, with increases ranging from 21.5% to 26.8%. In

TABLE IV
OPTIMAL PLANNING SCHEMES OF PV PROSUMERS IN CASE 2 AND 3

PV prosumer (node)	Case	Planning scheme (in unit)		Daily PV-grid-EV transactions amount (MW)
		PV	ES	
1(5)	2	16	7	3.758
	3	13	5	3.237
2(6)	2	15	8	3.579
	3	13	4	2.974
3(10)	2	13	6	3.092
	3	10	5	2.578
4(11)	2	10	4	2.634
	3	9	4	2.371
5(13)	2	4	2	0.879
	3	6	3	1.472
6(16)	2	4	2	0.961
	3	7	4	1.568
7(19)	2	11	5	2.833
	3	12	4	2.972
8(20)	2	6	3	1.489
	3	11	5	2.635
9(23)	2	5	3	1.194
	3	8	4	1.926
10(29)	2	3	1	0.837
	3	6	2	1.547

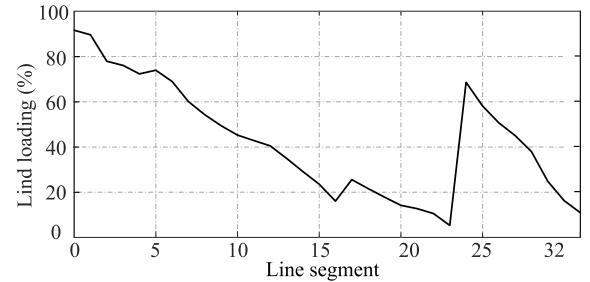


Fig. 4. Worst-case line loading, % of line capacity, %.

Case 3, EVCSs are not restricted to purchasing energy with IDSO at fixed TOU prices. Instead, they have the flexibility to exchange energy at a more competitive price in the transactive market to pursue higher revenues. Consequently, PV-grid-EV transactions are cost-efficient operational alternatives to the expansion planning of EVCSs and distribution networks.

2) *The impact of Network Charges on the Planning of EVCS and Distribution System by Comparing Cases 2 and 3:* As shown in Tables II and III, the disparity in total revenue of the IDSO and EVCSs in Case 2 and Case 3 is attributed to the absence of regulation in PV-grid-EV transactions in Case 2. In Case 3, IDSO experiences a significant 8.3% decrease in total costs due to the lesser additional capacity required for expanded substations and lines compared to Case 2. The total revenue of EVCSs in Case 3 has been improved to varying degrees compared to Case 2, with increases ranging from 8.7% to 16.1%. Additionally, there is a significant decrease in the average waiting time of EV users in Case 3 compared to Case 2. This improvement can be attributed to two factors.

Firstly, EVCSs and EV users in Case 2 lack visibility into the comprehensive operational status of the coupled distribution and transportation systems and, thus, do not follow physical constraints within the networks. The designed optimal network charges and EV charging prices point out potential congestion and extended charging waits at EVCSs, guiding EVCSs and EV users. The impact of EV charging prices on regulating EV charging flow is further analyzed later. Secondly, EVCSs prioritize profit-driven decisions, favoring trading at high selling prices and low buying prices without necessarily prioritizing grid-friendliness. From the perspective of EV users, the charging routes, waiting time, and charging costs are the major concerns in lowering their charging expenditures.

Given the impact of the distance of PV-grid-EV transactions on network charges, EVCSs tend to procure energy from nearby prosumers to minimize these charges. Consequently, the capacity of PV installations for nearby prosumers is often positively correlated with EVCS capacity. As indicated in Table IV, in Case 2, influenced by the uneven distribution of charging loads, nearby prosumers surrounding EVCS 1 and 2 have installed more PV generators to meet the charging demand at these locations. In contrast, prosumers 5, 6, 9, and 10, situated in more remote areas, have installed fewer PV generators, resulting in lower daily energy transactions.

Conversely, Case 3 explicitly co-optimizes EV charging flow and, thereby, PV-grid-EV transactions with grid operation and expansion planning through incentivizing optimal network charge and EV charging prices, factoring grid-related costs into EVCSs and EV users' utilities. They are incentivized to take beneficial actions for themselves and the grid; as demonstrated in Table III, the average charging waiting time in Case 3 witnesses a sharp decrease compared with Case 2. EVCSs are incentivized to alter EV charging patterns and engage in PV-grid-EV transactions to alleviate network congestion and defer DNEP. The unregulated transactive environment leads to insufficient operation flexibility, advancing DNEP and prompting higher charging costs.

3) *The impact of Charging Prices on EV Charging Flow by comparing Case 1 and Case 3:* We evaluate the impact of EV charging prices on regulating EV charging flow by comparing results from Case 1 and Case 3. The comparison of hourly EV charging prices in EVCSs is provided in Fig. 5. In contrast to the traditional uniform charging price adopted in Cases 1 and 2, the proposed optimal charging prices entail varying charging prices at different EVCSs. Specifically, in central areas with higher charging loads, EV users are subjected to relatively higher charging prices during rush hours to alleviate network pressure, as exemplified by EVCS 1 and 2 from 18:00 to 21:00. Conversely, in remote areas with lower charging loads, charging prices are comparatively lower to attract a broader range of EVs for charging, as observed at EVCS 4 and 5.

Fig. 6 further shows that the distribution of hourly EV charging flow at 5 EVCSs becomes more balanced in Case 3, especially in the rush hours (16:00-20:00). During rush hours, EV charging flows at EVCS 1 and 2 in Case 3 witness a significant decrease compared with Case 1. This is because we incorporate the spatial-temporal differential EV charging

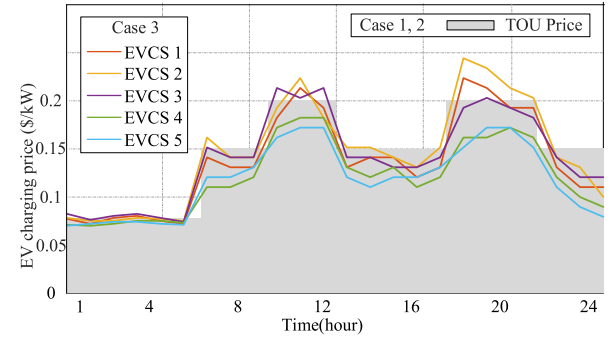


Fig. 5. Hourly EV charging price of EVCS in three Cases.

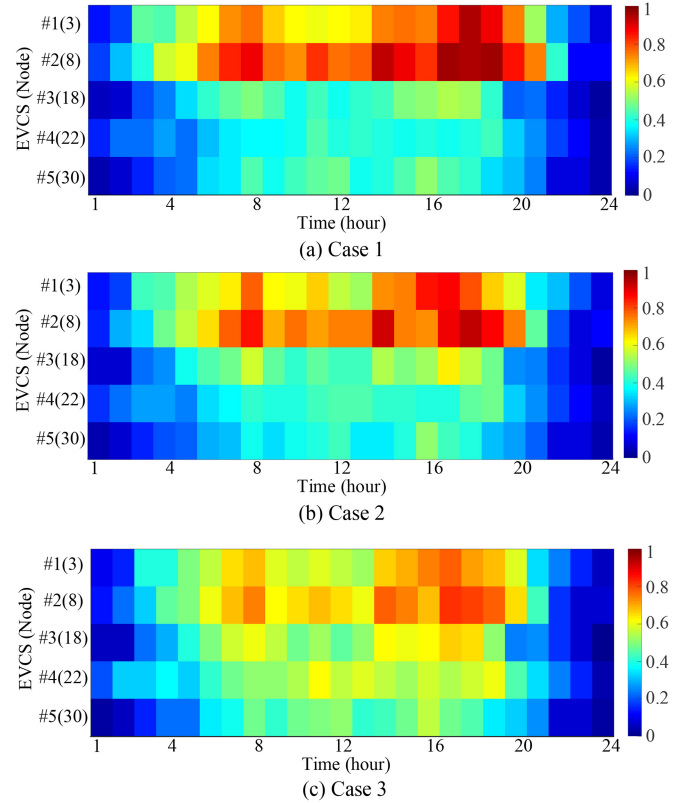


Fig. 6. Hourly EV charging flow at each EVCS, MWh.

prices in Case 3, which internalizes the global operation status of distribution and transportation networks. Specifically, in rush hours, the extremely high EV charging prices of EVCS 1 point out the potential occurrence of network congestion and charging wait at EVCS 1. In that case, EV users turn to charge at EVCS 4 and 5 with monetary incentives.

Fig. 7 compares the EV waiting time at EVCS 1 in Case 1 and Case 3 to evaluate the impact of optimal EV charging prices on improving EV users' satisfaction. It is observed that the 90% confidence level of the waiting time of Case 1 falls to around 6-24 minutes, while the 90% confidence level of Case 3 falls to around 5-20 minutes. It further demonstrates the effectiveness of optimal EV charging prices in regulating EV charging flow. EVCS leverages their decisions on resources in both the long-term and short-term to maximize its interests while simultaneously adhering to network securities.

TABLE V
OPTIMAL PLANNING SCHEMES AND COSTS OF EVCS, PV PROSUMERS, AND DISTRIBUTION NETWORKS IN THREE CASES

Case	Planning schemes		Investment costs for line and substation (10 ⁴ \$)	Operation costs (10 ⁴ \$)	Average waiting time (min)	Total computation time (s)
	EVCS (in chargers)	PV prosumer (in units)				
1	#1(9), #2(16), #3(7), #4(14), #5(10), #6(9)	/	328.2 (0%)	1164.89 (0%)	13.78 (0%)	1326
2	#1(8), #2(20), #3(5), #4(18), #5(14), #6(7)	#1(12), #2(4), #3(15), #4(16), #5(17)	269.4 (-17.9%)	982.01 (-15.7%)	12.09 (-12.3%)	1598
3	#1(11), #2(18), #3(9), #4(16), #5(13), #6(10)	#1(9), #2(6), #3(13), #4(14), #5(15)	246.5 (-24.5%)	891.14 (-23.5%)	10.87 (-21.1%)	4328

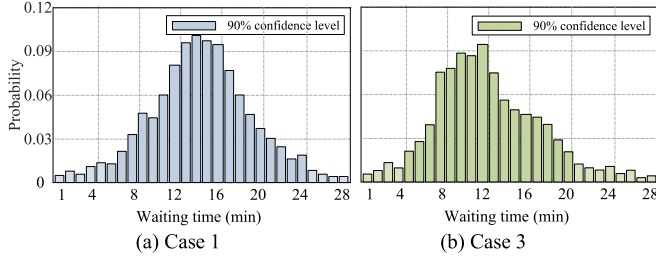


Fig. 7. The probability distribution of the EV waiting time.

C. Scalability Analysis Based on a Real-World Test System

To further demonstrate the scalability and effectiveness of the proposed method, we apply the model to a real-world zone within a city in central China. The initial topology of the coupled 72-node distribution system and 40-node transportation system is shown in Fig. 8. The transportation system is simplified from several primary districts within the city, comprising 40 nodes and 66 main roads. The power distribution system is represented by a 72-node system with 86 lines and four 110kV/10kV substations. The detailed parameters of the coupled real-world system are given in the Appendix [33].

The graphical topologies for the expanded distribution systems obtained for Case 1-3 are depicted in Fig. 9. Notably, in all cases, the meshed-constructed distribution system is operated radially, where four independent power supply zones are separated. For improved reliability, the four radial-operated zones are connected via normally open tie switches, allowing some portion of the faulted circuit to be restored quickly. The planning scheme and costs of EVCSs, PV prosumers and the distribution systems, as well as the total computation time, are summarized in Table V. Table V shows that the total computation time of Case 3 is longer than that of Cases 1 and 2. This is because Cases 1 and 2 neglect the PV-grid-EV transaction and the monetary incentive, with the IDSO directly determining DNEP. In contrast, the IDSO in Case 3 had to interact with EVCS and PV prosumers to enable multi-stakeholder collaborative planning, resulting in iterative optimization and, thus, longer computation times.

Results from Fig. 9 and Table V reveal that the expanded capacities of distribution lines and substations in Case 1 surpass those in Case 3, resulting in a notable decrease of 5.9×10^5 \$ in corresponding investment costs. This discrepancy

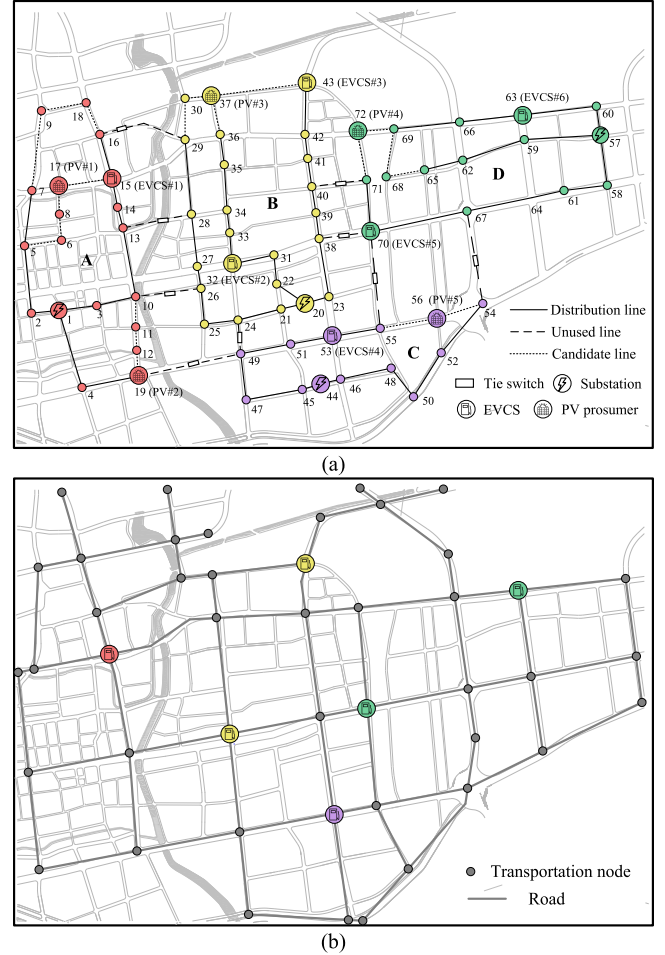


Fig. 8. Topology of the coupled (a) 72-node distribution system and (b) 40-node transportation system.

arises from the absence of PV installations in Case 1, necessitating all EV charging loads to be exclusively powered through the substations and several main distribution lines. Introducing PV-grid-EV transactions in Case 3 facilitates a more equitable distribution of EV charging and power flow, consequently deferring expansion planning and reducing operation costs. Additionally, more EV charging capacities are constructed in Case 3 compared to Case 1, suggesting that PV installations and PV-grid-EV transactions can mitigate electrical problems in the distribution networks, absorb uncertainties in EV charging patterns, and accommodate more charging loads.

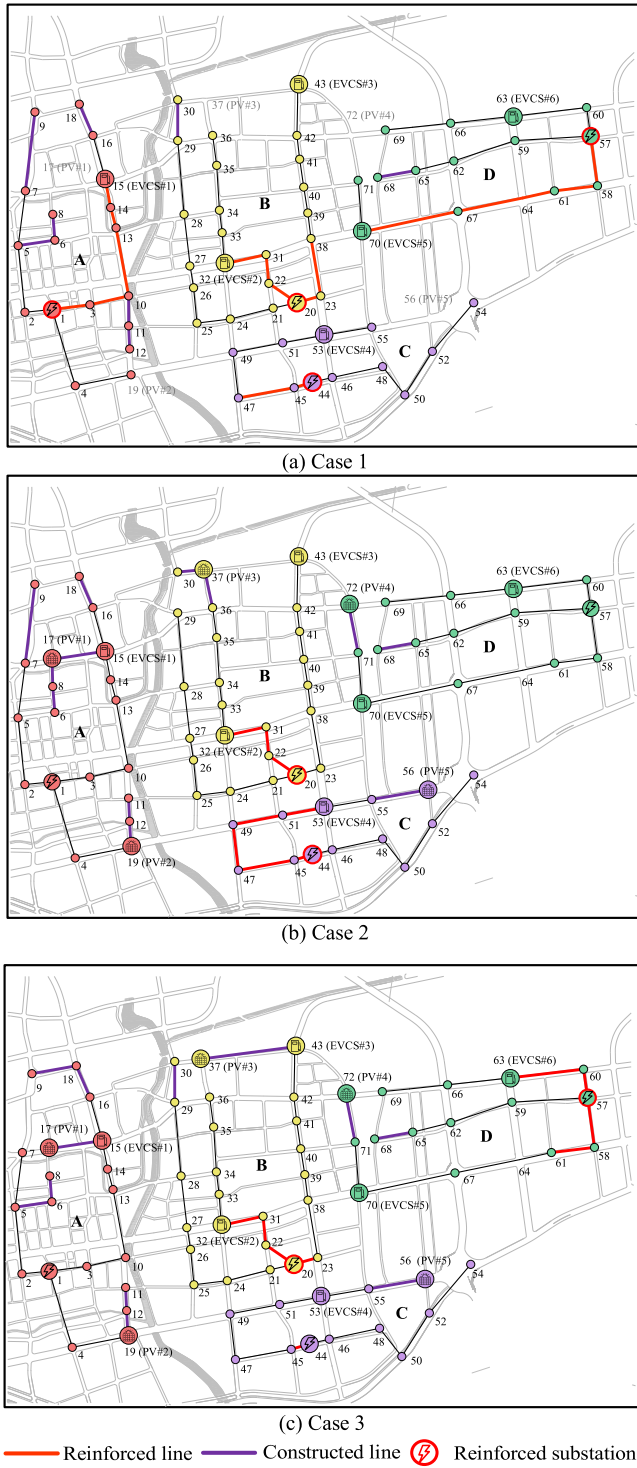


Fig. 9. Graphical topology of the final planning scheme of the distribution systems (a) Case 1; (b) Case 2; (c) Case 3.

Furthermore, comparing results between Case 2 and Case 3 reveals a decrease of $2.3 \times 10^5 \$$ in network investment costs. The geographical distribution of EV charging capacity is more balanced in Case 3 than in Case 2. Fig. 9(b) and (c) depict the differing network topologies between Case 2 and Case 3, particularly in constructing newly constructed lines connecting nodes 6, 9, 30, and 37. This disparity stems from the absence of optimal EV charging prices in Case 2, where charging

prices at all EVCSs are uniform for EV users. Consequently, their charging decisions remain independent of the distribution networks' operational status and planning schemes. EV users tend to opt for the shortest charging routes, directing more users to EVCSs in the central zone (i.e., EVCS 2, 4, and 5), thus increasing waiting time at EVCSs and exacerbating the burden on nearby distribution lines and substations. In contrast, implementing optimal EV charging prices in Case 3 allows for internalizing potential electrical issues in the congested zones, incentivizing EV users to choose remote EVCSs in less congested areas.

VII. CONCLUSION

This paper proposes a joint planning model of EVCS and distribution systems considering PV-grid-EV transactions to accommodate the large-scale integration of EV and PV more effectively. EVCSs and PV prosumers leverage their long-term and short-term resources to maximize their interests while respecting security constraints. We design a decentralized algorithm to preserve privacy and ensure convergence. The conclusions demonstrated by case studies are listed below:

PV-grid-EV transactions have proved to be a cost-efficient operational alternative to long-term planning. EVCSs and PV prosumers are not confined to autonomous planning to meet energy demand, leading to suboptimal utilities. It facilitates the joint planning of the networks, EVCSs, and PV prosumers by utilizing EV charging flexibility and other flexible sources.

The designed network charge and EV charging price are cost-reflective to incentivize EVCSs and EV users to align with grid-friendliness objectives. It serves as a fair incentive to coordinate planning decisions of different entities by factoring grid-related costs into the objectives based on their respective responsibilities. As for EV users, it points out potential network congestion and charging wait, replacing the extra dispatch on EV charging flow with energy transactions, thus reducing EV users' expenditure.

Future work could include the expansion planning of the transportation systems to further evaluate the impact of PV-grid-EV transactions on replacing the extra dispatch on EV charging flow and deferring planning of transportation systems.

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